

Patient Shortage in Histopathology: Training Against Class-Centre Uncertainty Gains 2.3 Points on TCGA-UT and Nothing Measurable on BRACS

Yannik Queisler

September 21, 2026

Abstract

A five-patient cohort estimates its class centres with an error whose covariance is the between-patient covariance divided by the patient count. Two attempts to estimate better centres from the cohort alone have not recovered the resulting headroom of 10.66 percentage points on TCGA-UT and 8.45 on BRACS. This report tests the complementary strategy: leave the centres as they are and train the classifier to be insensitive to their uncertainty. The loss is the implicit semantic data augmentation bound on the expected cross-entropy under Gaussian centre perturbations, with a perturbation covariance estimated from the cohort’s own patient means. On TCGA-UT, the arm with validation-tuned perturbation strength gains 2.34 points at five patients (95% interval: 2.03 to 2.63), meets the prespecified success criterion, recovers 22 percent of the pool-centre headroom, and narrows the five-to-twenty patient gap by 1.02 points (0.71 to 1.34). A trace-matched isotropic control gains 0.16 points (−0.01 to 0.31), so the gain depends on the patient directions. At five patients, the tuned arm exceeds directional shrinkage of the class centres by 1.06 points (0.81 to 1.30) and matches cohort whitening to within 0.08 points. Validation selects a perturbation 16 to 64 times the modelled centre variance in every TCGA-UT fit, inside the grid. On BRACS, the tuned arm gains 0.25 points (−0.37 to 0.97), which fails the criterion. The uncertainty loss and cohort whitening, two implementations of the same second-moment correction, thus reach the same TCGA-UT gain, above both centre-shrinkage remedies, and none recovers the BRACS headroom.

1 Motivation

Class-centre error carries most of the accuracy benefit of additional training patients. Shifting the class centres of a five-patient cohort to the centres of the full training pool gains 10.66 percentage points on TCGA-UT and 8.45 on BRACS (Queisler, 2026b,a). The shift needs the pool, so it measures headroom rather than offering a remedy. Two remedies that use only the drawn patients have tried to estimate better centres. Radial James-Stein shrinkage toward the grand mean recovers none of the TCGA-UT headroom (Queisler, 2026c). Directional shrinkage along the eigen-directions of the cohort’s pooled between-patient covariance gains 1.29 points on TCGA-UT and nothing measurable on BRACS, but it barely reduces centre error; its gain behaves like a first-moment form of cohort whitening (Queisler, 2026d).

These results suggest a change of target. A cohort cannot locate its class centres much better from its own patients, but it can estimate how uncertain they are. Under the patient model of Queisler (2026b), the mean of G patient means deviates from the class centre with covariance Σ/G , where Σ is the between-patient covariance. A classifier whose decisions change little when every class centre moves by such an error should lose less to the error it cannot remove. This report asks: *does training a linear classifier against the estimated class-centre uncertainty of its own cohort recover part of the pool-centre headroom?* The answer is judged at five patients per

class, separately on each dataset, against a practical threshold of one percentage point that was fixed before fitting.

Three strands of prior work supply the ingredients. Implicit semantic data augmentation (Wang et al., 2019) perturbs deep features along class-conditional covariance directions and shows that the expected cross-entropy under Gaussian perturbations has a closed-form upper bound, so the augmentation never has to be sampled. Learning with marginalized corrupted features (van der Maaten et al., 2013) makes the same move for linear models: it replaces explicit corrupted copies of the training data by the expected loss under the corruption distribution, which keeps fitting deterministic and cheap. MetaSAug (Li et al., 2021) observes that class covariances estimated from few samples are unreliable in long-tailed recognition, which motivates pooling the patient deviations across classes here and testing the covariance against a direction-free control. None of these works addresses patient shortage in histopathology; the adaptation below is proposed in this report.

2 Uncertainty model

Write the patch features of class c as $\mathbf{x}_{ik} = \boldsymbol{\mu}_c + \mathbf{u}_i + \mathbf{e}_{ik}$, with class centre $\boldsymbol{\mu}_c$, patient offset $\mathbf{u}_i$, and within-patient residual $\mathbf{e}_{ik}$ (Queisler, 2026b). A cohort of G patients per class estimates $\boldsymbol{\mu}_c$ by the mean $\hat{\boldsymbol{\mu}}_c$ of its patient means $\mathbf{m}_{ci}$. Pooling the deviations $\mathbf{m}_{ci} - \hat{\boldsymbol{\mu}}_c$ over all C classes at $C(G - 1)$ degrees of freedom gives the cohort’s between-patient covariance $\hat{\boldsymbol{\Sigma}}$, the same cohort basis used by cohort whitening (Queisler, 2026e) and directional shrinkage (Queisler, 2026d). Its rank is at most $C(G - 1)$. The *centre uncertainty* is $\mathbf{V}_G = \hat{\boldsymbol{\Sigma}}/G$.

Suppose every training feature of class y is displaced by a random centre error $\boldsymbol{\delta} \sim \mathcal{N}(\mathbf{0}, t\mathbf{V}_G)$. For a multinomial logistic model with logits $z_{ik} = \mathbf{w}_k^\top \mathbf{x}_i + b_k$, Jensen’s inequality bounds the expected cross-entropy of a patch by the loss inside the sum below (Wang et al., 2019). The classifier minimizes

$$\frac{1}{N} \sum_{i=1}^N \log \sum_{k=1}^C \exp \left[z_{ik} - z_{iy_i} + \frac{t}{2} (\mathbf{w}_k - \mathbf{w}_{y_i})^\top \mathbf{V}_G (\mathbf{w}_k - \mathbf{w}_{y_i}) \right] + \frac{\lambda}{2} \|\mathbf{W}\|_F^2. \quad (1)$$

The quadratic term raises the loss of every competing class k in proportion to how much the score difference between k and the true class varies when the class centre moves along the patient directions. The classifier can lower it by relying less on those directions for the separation of k from y_i . At $t = 0$, Equation 1 is ordinary multinomial logistic regression. The bound holds in expectation under the Gaussian model; it is not a worst-case guarantee, and it does not claim that the perturbation scale $1/G$ is correct for the test population. Prediction uses the ordinary logits on unchanged features.

The penalty is related to cohort whitening. Whitening all features by $(\mathbf{I} + \kappa \hat{\boldsymbol{\Sigma}})^{-1/2}$ and fitting a ridge penalty $\lambda \|\mathbf{W}'\|_F^2$ on the whitened features is the same as fitting the original features with the penalty $\lambda \sum_k \mathbf{w}_k^\top (\mathbf{I} + \kappa \hat{\boldsymbol{\Sigma}}) \mathbf{w}_k$. Both remedies therefore discourage weight along the cohort’s patient directions. They differ in where the penalty acts: whitening adds it once to the regularizer, whereas Equation 1 adds it to each training patch inside the softmax, where it matters most for patches whose competing classes are close.

3 Design

The study keeps the setting of Queisler (2026b) on TCGA-UT and of Queisler (2026a) on BRACS. TCGA-UT contributes 30 cancer types and BRACS its seven lesion classes, both with frozen 2,560-dimensional Virchow2 features (Zimmermann et al., 2024), three fixed patient-disjoint splits, and multinomial logistic regression whose regularization strength is selected on validation patient-macro balanced accuracy. For each split and each of ten draws, twenty eligible

patients per class are sampled, and nested cohorts of five patients with 32 patches, ten with sixteen, and twenty with eight are formed, so every classifier trains on 160 patches per class at a single patient count for all classes. No additional patients, labels, features, or foundation models enter.

Three new arm families enter at each $G \in \{5, 10, 20\}$ (Table 1). All three minimize Equation 1 on the unchanged training features and differ only in the covariance and in how t is set. The isotropic control replaces $\mathbf{V}_G$ by $\text{tr}(\mathbf{V}_G)\mathbf{I}/2,560$: it keeps the total perturbation variance but spreads it evenly over all feature dimensions, so it tests whether the patient directions matter or only the amount of perturbation. The covariance is estimated from the training cohort alone; validation and test features never enter it. The reused arms are the exact fits of the source studies on the same cohorts, so every contrast is paired draw by draw.

Arm	Source	Training objective
RG	reused	Ordinary logistic regression on the real cohort of G patients
CG	reused	Real cohort with class centres shifted to the pool centres (oracle reference)
SG, StG	reused	Radial centre shrinkage, closed form and tuned (Queisler, 2026c)
AG, AtG	reused	Directional centre shrinkage, closed form and tuned (Queisler, 2026d)
$RWcG$	reused	Cohort-basis whitening (Queisler, 2026e,a)
UG	new	Equation 1 with patient covariance $\mathbf{V}_G$ at $t = 1$
UtG	new	Equation 1 with patient covariance, t selected from $\{0, 0.25, 1, 4, 16, 64, 256\}$
ItG	new	Equation 1 with isotropic covariance $\text{tr}(\mathbf{V}_G)\mathbf{I}/2,560$, same grid

Table 1: The three new families use only the drawn patients and change the loss, not the features. RG and CG bracket the available headroom; S, St, A, At, and RWc are the cohort-only remedies already tested. The $t = 0$ candidate of Ut and It is the RG fit itself.

Each new candidate is fitted with L-BFGS-B in double precision with an analytic gradient, the unpenalized intercepts, tolerance, and iteration limit of the ordinary logistic fits, and the regularization grid $\lambda \in \{10^{-6}, \dots, 10^2\}$ of nine values. The quadratic forms are evaluated through the low-rank eigenbasis of $\hat{\Sigma}$, so no $2,560 \times 2,560$ covariance is formed. The tuned families select t jointly with λ on validation patient-macro balanced accuracy, with ties resolved toward the smaller t and then the larger λ ; the fixed family U selects only λ among the $t = 1$ candidates. Selections were written to disk before any test prediction, and test accuracy never entered selection. Per dataset, the study fits $2 \times 6 \times 9$ candidates for each of three patient counts and 30 split-draw pairs, 9,720 candidates and 19,440 in total. The grid was frozen before fitting, and test results never informed it.

4 Evaluation

The endpoint is patient-macro balanced accuracy on the test partition, pooled over splits and draws with equal weight per split. Uncertainty combines Bayesian bootstrap weights on test patients (Rubin, 1981) with resampling of draws within each split over 2,000 replicates; all arms share the replicates, so every contrast is paired, and intervals are 95 percent percentile intervals.

The primary contrast is Ut5–R5, read separately on each dataset. The remedy counts as a practical success on a dataset if the estimated gain is at least one percentage point and its paired interval lies entirely above zero; a claim across both cohorts requires success on both. Two further outcomes are distinguished. A *patient-direction benefit* requires Ut to exceed the isotropic control It. A *stronger existing remedy* requires Ut to exceed cohort whitening RWc; otherwise whitening is reported as competitive or stronger. Secondary quantities are the gains at ten and twenty

patients, the contrasts Ut–At and Ut–St, the *recovery* $\rho_G = (\text{Ut}G - \text{RG})/(\text{CG} - \text{RG})$ of the pool-centre headroom, and the *gap reduction* $\Delta^R - \Delta^{\text{Ut}}$ of the patient-count gap $\Delta^X = XG' - XG$ together with its share of Δ^R . A gap reduction is claimed only when its paired interval excludes zero. Recovery and shares are computed per replicate. The headroom intervals exclude zero on both datasets at every patient count, so no recovery ratio has a denominator whose interval includes zero.

5 Results

Training against centre uncertainty meets the prespecified success criterion on TCGA-UT and fails it on BRACS. On TCGA-UT, the tuned arm gains 2.34 points at five patients (95% interval: 2.03 to 2.63), the fixed arm 0.52 points (0.44 to 0.59), and both gains shrink with the patient count, to 1.32 and 0.13 points at twenty (Table 2). On BRACS, the tuned arm gains 0.25 points (−0.37 to 0.97) and the fixed arm 0.20 (−0.12 to 0.70). The primary contrast therefore succeeds on one dataset only, and no claim across both cohorts is supported. The isotropic control stays below 0.25 points on both datasets at every patient count, so it accounts for at most a tenth of the tuned gain on TCGA-UT.

Arm family	$G = 5$	$G = 10$	$G = 20$
<i>TCGA-UT</i>			
Real (R)	62.48 [61.24, 63.65]	67.79 [66.47, 69.00]	71.89 [70.55, 73.06]
Patient covariance, $t = 1$ (U)	63.00 [61.77, 64.17]	68.04 [66.73, 69.24]	72.02 [70.69, 73.20]
Patient covariance, tuned (Ut)	64.82 [63.60, 66.00]	69.56 [68.24, 70.75]	73.21 [71.93, 74.36]
Isotropic control (It)	62.64 [61.39, 63.82]	68.00 [66.69, 69.20]	71.98 [70.63, 73.17]
Pool centres (C)	73.14 [71.62, 74.50]	74.47 [73.01, 75.78]	75.55 [74.11, 76.85]
Gain U – R	0.52 [0.44, 0.59]	0.25 [0.19, 0.31]	0.13 [0.12, 0.15]
Gain Ut – R	2.34 [2.03, 2.63]	1.77 [1.48, 2.05]	1.32 [1.08, 1.55]
Gain It – R	0.16 [−0.01, 0.31]	0.21 [0.11, 0.32]	0.09 [0.01, 0.17]
Headroom C – R	10.66 [9.78, 11.47]	6.68 [6.07, 7.28]	3.66 [3.27, 4.06]
<i>BRACS</i>			
Real (R)	36.18 [33.46, 38.79]	38.49 [35.87, 41.22]	40.97 [38.50, 43.62]
Patient covariance, $t = 1$ (U)	36.38 [33.55, 39.13]	38.46 [35.79, 41.18]	40.96 [38.51, 43.59]
Patient covariance, tuned (Ut)	36.43 [33.83, 39.04]	38.72 [36.06, 41.48]	41.31 [38.88, 43.91]
Isotropic control (It)	36.27 [33.55, 38.94]	38.48 [35.79, 41.26]	40.83 [38.41, 43.45]
Pool centres (C)	44.63 [41.33, 48.05]	45.11 [41.85, 48.51]	45.50 [42.20, 48.76]
Gain U – R	0.20 [−0.12, 0.70]	−0.04 [−0.42, 0.20]	−0.01 [−0.37, 0.17]
Gain Ut – R	0.25 [−0.37, 0.97]	0.23 [−0.30, 0.76]	0.34 [−0.14, 0.91]
Gain It – R	0.10 [−0.20, 0.39]	−0.02 [−0.41, 0.28]	−0.14 [−0.61, 0.12]
Headroom C – R	8.45 [6.31, 10.84]	6.61 [4.85, 8.58]	4.53 [2.77, 6.37]

Table 2: The patient-covariance loss gains about 2.3 points at five TCGA-UT patients and nothing distinguishable from zero on BRACS; the trace-matched isotropic control gains a fraction of that on either dataset. Accuracy entries are test patient-macro balanced accuracy in percent; gains and headroom are paired contrasts in percentage points; bold entries are the primary contrast. Intervals are 95% bootstrap intervals over test patients and draws. The R and C rows are those of Queisler (2026b) and Queisler (2026a).

Unlike the directional-shrinkage result (Queisler, 2026d), the TCGA-UT success does not hinge on the interval bound: the lower bound of 2.03 points lies twice the threshold above zero. The

gain is nonetheless small against the headroom of 10.66 points. No BRACS interval excludes zero at any patient count, so that dataset supports neither a gain nor a loss.

5.1 Headroom recovery and the patient-count gap

On TCGA-UT, the tuned arm recovers 0.220 (0.191 to 0.251) of the pool-centre headroom at five patients and 0.265 (0.222 to 0.313) at ten; the fixed arm recovers 0.048 and 0.037 (Table 3). Because the gain falls with the patient count, the tuned arm narrows the five-to-twenty gap from 9.41 to 8.39 points, a paired reduction of 1.02 points (0.71 to 1.34) and a share of 0.109 (0.077 to 0.140). Both single steps show reductions with intervals above zero (Figure 1). On BRACS, recovery is 0.030 (−0.050 to 0.106) at five patients and 0.034 at ten, and the gap reduction of −0.09 points (−0.84 to 0.70) is indistinguishable from zero. The isotropic control reduces no TCGA-UT gap by more than 0.12 points and no BRACS gap by more than 0.24 points, with intervals that include zero except for the ten-to-twenty step on TCGA-UT.

	TCGA-UT	BRACS
Recovery ρ_5, U	0.048 [0.041, 0.057]	0.024 [−0.013, 0.082]
Recovery ρ_5, Ut	0.220 [0.191, 0.251]	0.030 [−0.050, 0.106]
Recovery ρ_{10}, U	0.037 [0.027, 0.047]	−0.006 [−0.069, 0.029]
Recovery ρ_{10}, Ut	0.265 [0.222, 0.313]	0.034 [−0.049, 0.116]
Gap, real $\Delta_{5 \rightarrow 20}^R$	9.41 [8.71, 10.12]	4.80 [2.88, 6.75]
Gap, tuned $\Delta_{5 \rightarrow 20}^{Ut}$	8.39 [7.73, 9.05]	4.88 [3.01, 6.74]
Gap reduction, U, 5 $\rightarrow$ 20	0.38 [0.30, 0.46]	0.21 [−0.16, 0.75]
Gap reduction, Ut, 5 $\rightarrow$ 10	0.57 [0.30, 0.85]	0.02 [−0.71, 0.84]
Gap reduction, Ut, 10 $\rightarrow$ 20	0.45 [0.21, 0.68]	−0.11 [−0.80, 0.54]
Gap reduction, Ut, 5 $\rightarrow$ 20	1.02 [0.71, 1.34]	−0.09 [−0.84, 0.70]
Share, Ut, 5 $\rightarrow$ 20	0.109 [0.077, 0.140]	−0.018 [−0.213, 0.131]

Table 3: On TCGA-UT the tuned uncertainty loss captures a fifth of the pool-centre headroom and removes a ninth of the patient-count gap; on BRACS neither quantity is distinguishable from zero. Recovery is the gain divided by the headroom at the same patient count; the gap reduction is the paired difference of the real and the trained patient-count gaps, in percentage points, and the share divides it by the real gap.

5.2 Comparison with the control and the other cohort-only remedies

On TCGA-UT, the patient directions carry almost the entire gain. Ut exceeds the isotropic control by 2.18 points at five patients (1.89 to 2.49), 1.56 at ten, and 1.23 at twenty, all with intervals above zero. On BRACS the same contrast is 0.16 points (−0.56 to 0.98) at five patients and 0.48 (−0.06 to 1.12) at twenty, so no patient-direction benefit is supported there.

The ranking against the reused remedies separates the two families of cohort-only remedy (Figure 2). On TCGA-UT, Ut exceeds both centre-shrinkage arms at every patient count: the contrast Ut−At is 1.06 points at five patients (0.81 to 1.30), 1.07 at ten, and 0.91 at twenty, and Ut−St is 2.20 (1.91 to 2.49), 1.64, and 1.18 points. Against cohort whitening the two arms coincide: Ut−RWc is 0.08 points at five patients (−0.01 to 0.18), −0.08 at ten, and 0.05 at twenty, so every interval includes zero and none reaches a tenth of a point in either direction. On BRACS, Ut is indistinguishable from At, RWc, and St at every patient count, with point contrasts between −0.73 and 0.48 points; the widest of them, −0.73 against tuned radial shrinkage at five patients, has an interval from −1.64 to 0.17.

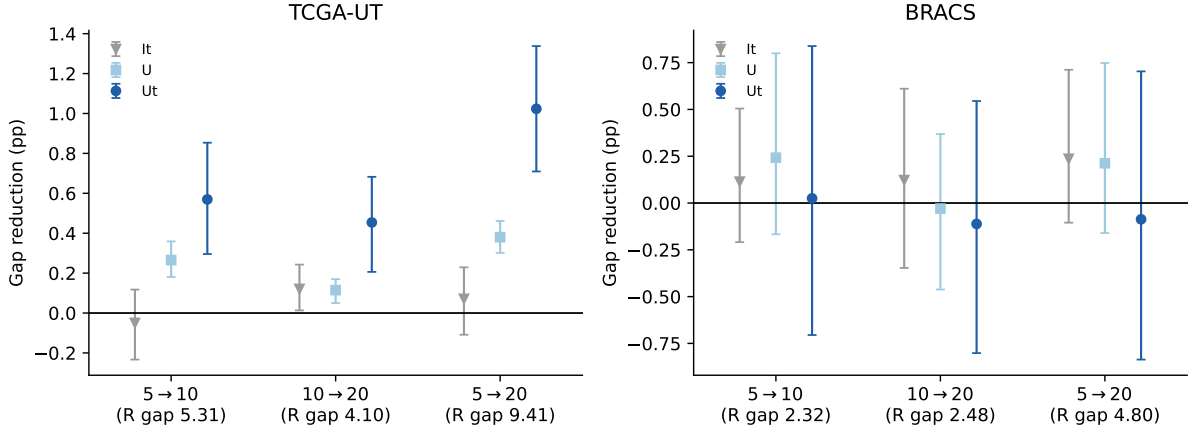


Figure 1: Only the patient-covariance loss narrows the patient-count gap, and only on TCGA-UT. Paired reduction of the real patient-count gap in percentage points, for each step between patient counts; the real gap is given below each step. Error bars show the paired 95% bootstrap intervals.

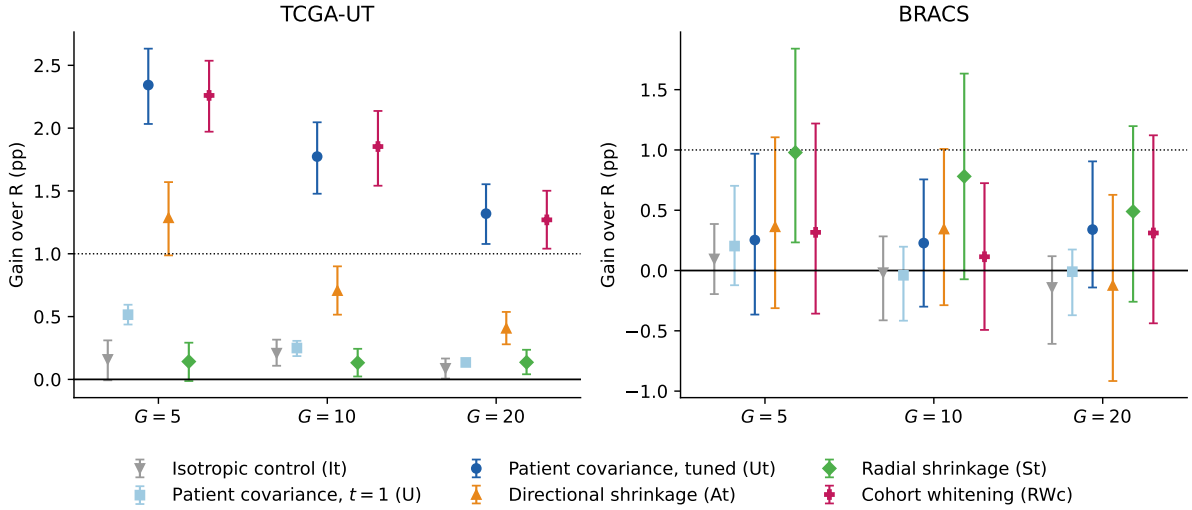


Figure 2: Centre-uncertainty training matches cohort whitening on TCGA-UT and exceeds both centre-shrinkage remedies; on BRACS all cohort-only remedies are indistinguishable from zero except radial shrinkage at five patients. Paired gains over the real cohort in percentage points; error bars show the paired 95% bootstrap intervals. The dotted line marks the one-point threshold.

5.3 Selected strengths and solver behaviour

Validation selection behaves differently on the two datasets (Table 4). On TCGA-UT, the tuned patient arm selected $t = 16$ or $t = 64$ in every one of the 90 fits and never reached the largest grid value, so the selection lies inside the grid at 16 to 64 times the modelled centre variance. The fixed arm at $t = 1$ gains less than a quarter as much as the tuned arm at five patients (Ut–U: 1.83 points, 1.55 to 2.09), so the modelled scale $\mathbf{V}_G$ is far below the strength that validation prefers. On BRACS, the tuned patient arm chose $t = 0$, the ordinary fit, in 2 to 6 of 30 fits per patient count and spread the remaining selections over the whole grid. The isotropic control chose the largest value $t = 256$ in 13 to 27 of 30 fits on both datasets, yet its accuracy never moved from R by more than 0.21 points.

Dataset	G	Rank	Arm	Selected t						
				0	0.25	1	4	16	64	256
TCGA-UT	5	120	Ut	0	0	0	0	10	20	0
			It	2	0	1	0	1	8	18
	10	270	Ut	0	0	0	0	10	20	0
			It	2	1	0	0	2	2	23
	20	570	Ut	0	0	0	0	0	30	0
			It	0	0	0	0	0	3	27
BRACS	5	28	Ut	6	6	3	2	3	4	6
			It	3	0	4	2	4	4	13
	10	63	Ut	4	4	4	2	11	4	1
			It	3	1	0	3	6	4	13
	20	133	Ut	2	2	2	7	4	8	5
			It	1	0	0	0	4	8	17

Table 4: TCGA-UT validation data favour a perturbation 16 to 64 times the modelled centre variance and never the largest grid value; BRACS validation data are divided, and the direction-free control drifts to the largest value on both datasets. Counts are tuned classifiers out of 30 per dataset and patient count (three splits, ten draws). The rank is that of the cohort’s pooled between-patient covariance, which equals its maximum $C(G - 1)$ in every draw.

The regularization strength reached a grid boundary in a minority of fits. On TCGA-UT, the fixed arm chose the smallest value $\lambda = 10^{-6}$ in 3 of 30 fits at five patients and 1 at ten, and the isotropic control chose it once at five patients; no tuned patient fit chose a boundary value, all of them selecting $\lambda \in \{0.1, 1\}$. On BRACS, the largest value $\lambda = 100$ was chosen by the fixed arm in 4 and 3 of 30 fits at five and ten patients, by the tuned patient arm in 8, 3, and 3 fits across the three patient counts, and by the isotropic control in 1 and 2 fits at five and ten patients.

Solver behaviour separates the two covariances. Every patient-covariance candidate converged on both datasets. Among the 9,720 candidates per dataset, 538 isotropic TCGA-UT candidates and 256 isotropic BRACS candidates stopped at the iteration limit of 10,000, all of them at $t \geq 16$ on TCGA-UT (61 at $t = 16$, 271 at $t = 64$, 206 at $t = 256$) and at $t \in \{4, 16, 64\}$ on BRACS (3, 239, and 14 candidates). Their gradient norms are small, between 2.4×10^{-8} and 1.6×10^{-4} on TCGA-UT and between 8.7×10^{-8} and 5.4×10^{-5} on BRACS, but the selection rule excludes every candidate that did not converge, so no arm reports one. No cohort produced a zero covariance.

5.4 Why the isotropic control gains little

The isotropic control has the same total perturbation variance as the patient covariance, but it spreads that variance over 2,560 dimensions instead of the 28 to 570 dimensions the cohort covariance spans. Its variance per dimension is therefore $r/2,560$ times the mean eigenvalue of $\mathbf{V}_G$, where r is the rank: 4.7 percent at five TCGA-UT patients and 1.1 percent at five BRACS patients. With $\mathbf{V}_G$ replaced by $\sigma^2 \mathbf{I}$, the quadratic term in Equation 1 becomes $\frac{t\sigma^2}{2} \|\mathbf{w}_k - \mathbf{w}_{y_i}\|^2$, a uniform penalty on weight differences that acts much like additional ridge regularization. The regularization grid can absorb such a penalty, which is consistent with the control gaining at most 0.21 points although it usually selects the largest strength in the grid, where its total perturbation variance is 256 times the modelled one. The control thus shows that uncertainty of the same total size, without direction, buys at most a tenth of what the patient directions buy on TCGA-UT. It does not show that any directional perturbation would help: a control matched per direction, for example random directions of the same rank, was not run.

6 Discussion

Training against the estimated uncertainty of the class centres clears the practical threshold on TCGA-UT: 2.34 points at five patients, with an interval from 2.03 to 2.63, a recovery of 22 percent of the pool-centre headroom, and a paired reduction of the five-to-twenty patient gap by 1.02 points. The gain depends on the patient directions, since a trace-matched isotropic perturbation recovers at most a tenth of it. On BRACS the remedy recovers nothing measurable at any patient count. Of the three outcomes defined in Section 4, TCGA-UT shows a practical remedy and a patient-direction benefit but no remedy stronger than cohort whitening, and BRACS shows none of the three.

The most informative result is the coincidence with cohort whitening. On TCGA-UT the two arms differ by at most 0.08 points at every patient count, although whitening transforms the features before an ordinary fit and the uncertainty loss leaves the features untouched and changes the objective. Section 2 shows why: whitening by $(\mathbf{I} + \kappa \hat{\Sigma})^{-1/2}$ with a ridge penalty is the same as an unwhitened fit whose penalty grows along the cohort’s patient directions, and Equation 1 adds a penalty of that form inside the softmax. The two remedies are therefore two implementations of the same second-moment correction, and they reach the same accuracy. Both exceed directional shrinkage by about one point at five patients and radial shrinkage by more than two, and shrinkage of the centres is a first-moment correction that leaves the cohort’s patient directions as trusted as before. On this reading, what a small cohort contributes is information about which directions not to trust rather than a better estimate of where its classes are, and acting on that information through the loss or through the features makes no measurable difference.

The strength that validation selects is the second informative result. Every TCGA-UT selection sits at 16 or 64 times $\mathbf{V}_G$, and the arm fixed at the modelled scale $t = 1$ gains 0.52 points against the tuned arm’s 2.34. The Gaussian model of Section 2 therefore gets the useful direction structure right and the useful scale wrong by more than an order of magnitude. Two mechanisms are consistent with this. The cohort covariance is estimated from $C(G - 1)$ deviations and is zero outside its span, so a large t compensates for the directions it misses; and the bound in Equation 1 is loose, so the penalty it imposes at the modelled scale is weaker than the expected loss it stands in for. The fits do not separate these, and the selected strength has no interpretation as a calibrated variance.

All cohort-only remedies built on the same patient covariance leave most of the headroom unrecovered. On TCGA-UT, the uncertainty loss and cohort whitening recover about a fifth of it at five patients, directional shrinkage about an eighth, and radial shrinkage nothing measurable; on BRACS none recovers a measurable part. The remaining eight to nine points at five patients

are not reached by estimating centres better from the cohort, nor by training against their uncertainty. This strengthens the conclusion of the preceding reports that the missing centre information must come from outside the labelled cohort.

7 Limitations

The uncertainty model makes three assumptions that the study does not verify. It pools the between-patient covariance across classes, so all classes are assumed to share the same patient-level noise. It uses the covariance of the cohort’s own patient means, which includes the finite-patch noise of each patient mean, so $\mathbf{V}_G$ slightly overstates the between-patient part. And it scales that covariance by $1/G$ and treats the centre error as Gaussian. The covariance is estimated from $C(G-1)$ deviations in 2,560 dimensions, only 28 for BRACS at five patients, and is zero outside its span, so most of the feature space receives no perturbation. The accuracy gain does not show that the perturbation matches the true centre error, and the selected strengths of 16 to 64 times the modelled variance show that the $1/G$ scaling does not predict what validation prefers.

The strength grid spans $t \leq 256$, which the tuned patient arm never selected on TCGA-UT and selected in 6, 1, and 5 of 30 fits on BRACS, so the reported gains are not bounded by the grid on TCGA-UT. The direction-free control does drift to the largest value, so its reported effect is a grid-bounded one. The isotropic control matches only the total variance; it does not separate the effect of the specific patient directions from that of any low-rank perturbation of the same size. All arms reuse the three evaluation splits of the source studies, so they share test patients with every earlier comparison in this series, and repeated use of these partitions across studies inflates the chance that some remedy appears to work by selection alone. The results are conditional on frozen Virchow2 features, multinomial logistic regression, 160 patches per class, and the two cohorts studied; transfer to other cohorts, feature extractors, or class counts is untested.

References

- Shuang Li, Kaixiong Gong, Chi Harold Liu, Yulin Wang, Feng Qiao, and Xinjing Cheng. MetaSAug: Meta semantic augmentation for long-tailed visual recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5212–5221, 2021.
- Yannik Queisler. Patient shortage in histopathology: Centre error carries the patient benefit on bracs, but the whitening channel does not separate. Technical report, TU Berlin, 2026a. Companion BRACS mechanism report, Experiment 18.
- Yannik Queisler. Patient shortage in histopathology: Class-centre error explains most of the benefit of additional patients. Technical report, TU Berlin, 2026b. Companion centre-error report, Experiment 16.
- Yannik Queisler. Patient shortage in histopathology: Shrinking class centres toward the cohort’s grand mean recovers almost none of the centre-error headroom. Technical report, TU Berlin, 2026c. Centre-shrinkage report, Experiment 19.
- Yannik Queisler. Patient shortage in histopathology: Directional shrinkage of class centres gains about one point on tcga-ut and nothing measurable on bracs. Technical report, TU Berlin, 2026d. Directional-shrinkage report, Experiment 20.

- Yannik Queisler. Patient shortage in histopathology: Whitening along between-patient directions is a separate, additive channel. Technical report, TU Berlin, 2026e. Companion patient-directions report, Experiment 17.
- Donald B. Rubin. The bayesian bootstrap. *The Annals of Statistics*, 9(1):130–134, 1981. doi: 10.1214/aos/1176345338.
- Laurens van der Maaten, Minmin Chen, Stephen Tyree, and Kilian Q. Weinberger. Learning with marginalized corrupted features. In *Proceedings of the 30th International Conference on Machine Learning*, volume 28 of *Proceedings of Machine Learning Research*, pages 410–418, 2013.
- Yulin Wang, Xuran Pan, Shiji Song, Hong Zhang, Cheng Wu, and Gao Huang. Implicit semantic data augmentation for deep networks. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- Eric Zimmermann, Eugene Vorontsov, Julian Viret, Adam Casson, Michal Zelechowski, George Shaikovski, Neil Tenenholtz, James Hall, Thomas Fuchs, Nicolo Fusi, Siqi Liu, and Kristen Severson. Virchow2: Scaling self-supervised mixed magnification models in pathology. *arXiv preprint arXiv:2408.00738*, 2024. doi: 10.48550/arXiv.2408.00738.